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Project-Specific Performance Specification for “CAMPO LINDO, CL” Project V150-4.3 MW Mk3E



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This Project-Specific Performance Specification is valid solely for the "CAMPO LINDO, CL" project specified in ISM 20-0045.

See general reservations, notes and disclaimers (including, Section 4, p. 6) to this Performance Specification.

1 General Description

The standard Vestas V150-4.0 MW wind turbine is able to be operated in Power Optimized (PO) modes.

This Performance Specification contains preliminary power curves and C_t curves for the following combinations of Power Optimized mode:

- Power Optimized Mode PO2/PO2-0S – 4.3MW

2 Operational Envelope and Performance Guidelines

Actual climate and site conditions have many variables and should be considered in evaluating actual turbine performance. The design and operating parameters set forth in this section do not constitute warranties, guarantees, or representations as to turbine performance at actual sites.

2.1 Operational Envelope – Conditions for Power Curve and C_t Values (at Hub Height)

Consult Section 5, p. 7 for power curves and C_t values.

Conditions for Power Curve and C_t Values (at Hub Height)	
Wind Shear, α	0.00-0.30 (10 minute average)
Turbulence Intensity, I	6-12% (10 minute average)
Blades	Clean
Rain	No
Ice/Snow on Blades	No
Leading Edge	No damage
Terrain	IEC 61400-12-1
Inflow Angle (Vertical)	$0 \pm 2^\circ$
Grid Voltage	Nominal Voltage $\pm 2.5\%$
Grid Frequency	Nominal Frequency ± 0.5 Hz
Grid Active Power (at LV-side of turbine transformer)	As per tabulated values in Section 5.
Grid Reactive Power (at LV-side of turbine transformer)	Power Factor 1.0

Table 2-1: Conditions for power curve and C_t values

3 Drawings

3.1 Structural Design – Illustration of Outer Dimensions

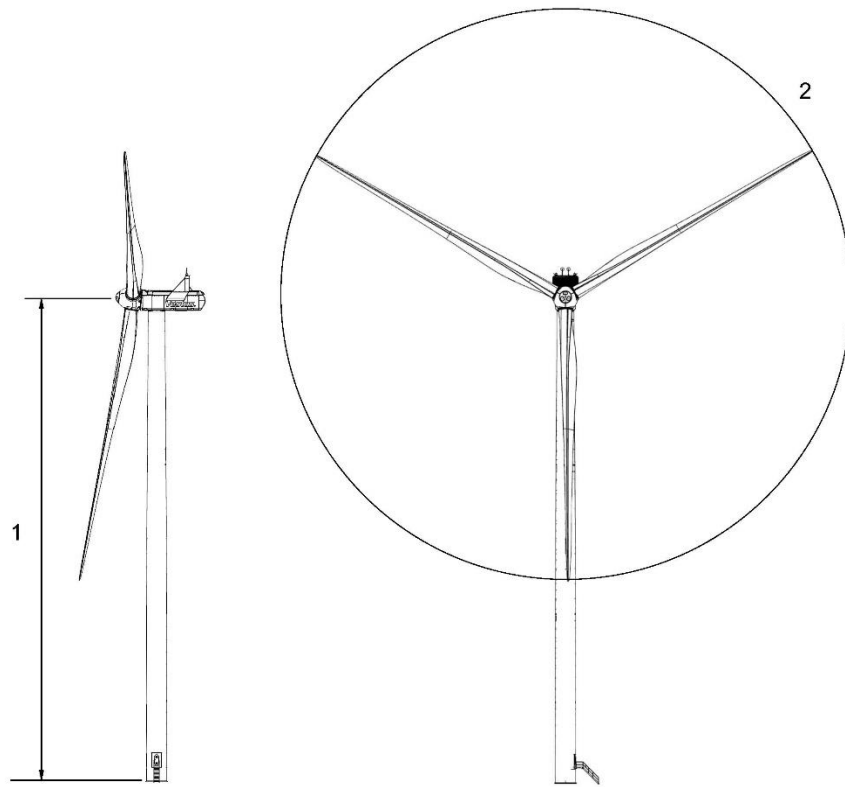


Figure 3-1: Illustration of outer dimensions – structure

1 Hub height: 140 m

2 Diameter: 150 m

4 General Reservations, Notes and Disclaimers

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- The performance specifications described in this document apply to the current version of the V150-4.0 MW wind turbine. Updated versions of the V150-4.0 MW wind turbine, which may be manufactured in the future, may differ from these performance specifications. In the event that Vestas supplies an updated version of the V150-4.0 MW wind, Vestas will provide an updated performance specification applicable to the updated version.
- All listed start/stop parameters (e.g. wind speeds) are equipped with hysteresis control. This can, in certain borderline situations, result in turbine stops even though the ambient conditions are within the listed operation parameters.
- This document, Performance Specification, is not an offer for sale, and does not contain any guarantee, warranty and/or verification of the power curve and sound (including, without limitation, the power curve and sound verification method). Any guarantee, warranty and/or verification of the power curve and sound (including, without limitation, the power curve and sound verification method) must be agreed to separately in writing.

5 Power Curves, Ct Values & Sound Power Curves

5.1 Power Curves, Power Optimized Mode PO2/PO2-0S

Wind	Air density kg/m ³													
	1.225	0.950	0.975	1.000	1.025	1.050	1.075	1.100	1.125	1.150	1.175	1.200	1.250	1.275
3.0	81	51	54	56	59	62	65	67	70	73	76	78	84	86
3.5	172	122	127	131	136	140	145	149	154	158	163	167	176	181
4.0	285	210	217	223	230	237	244	251	258	264	271	278	291	298
4.5	424	317	327	337	346	356	366	375	385	395	404	414	433	443
5.0	596	451	464	477	491	504	517	530	543	557	570	583	609	622
5.5	808	615	632	650	668	685	703	720	738	755	773	790	825	843
6.0	1061	811	834	857	880	902	925	948	971	993	1016	1039	1084	1106
6.5	1360	1044	1072	1101	1130	1159	1188	1216	1245	1274	1303	1331	1389	1418
7.0	1711	1316	1353	1389	1425	1461	1496	1532	1568	1604	1640	1675	1746	1781
7.5	2106	1629	1673	1717	1761	1805	1848	1891	1935	1978	2021	2064	2149	2191
8.0	2550	1982	2034	2086	2138	2191	2242	2294	2346	2397	2448	2499	2601	2651
8.5	3046	2378	2439	2501	2563	2625	2686	2746	2807	2868	2927	2987	3104	3162
9.0	3554	2812	2883	2953	3024	3095	3162	3229	3296	3364	3427	3491	3613	3671
9.5	3941	3249	3323	3397	3471	3545	3607	3669	3732	3794	3843	3892	3981	4020
10.0	4170	3659	3723	3787	3850	3914	3958	4001	4044	4088	4115	4143	4188	4206
10.5	4256	3977	4019	4061	4102	4144	4165	4186	4206	4227	4236	4246	4262	4269
11.0	4296	4189	4207	4226	4244	4262	4269	4275	4282	4288	4291	4293	4297	4298
11.5	4300	4264	4271	4278	4285	4292	4293	4295	4297	4299	4299	4300	4300	4300
12.0	4300	4290	4292	4294	4297	4299	4300	4300	4300	4300	4300	4300	4300	4300
12.5	4300	4298	4299	4299	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
13.0	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
13.5	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
14.0	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
14.5	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
15.0	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
15.5	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
16.0	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
16.5	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
17.0	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300	4300
17.5	4296	4296	4296	4296	4296	4296	4296	4296	4296	4296	4296	4296	4296	4296
18.0	4230	4230	4230	4230	4230	4230	4230	4230	4230	4230	4230	4230	4230	4230
18.5	4102	4102	4102	4102	4102	4102	4102	4102	4102	4102	4102	4102	4102	4102
19.0	4003	4002	4002	4002	4002	4002	4002	4002	4002	4002	4003	4003	4003	4003
19.5	3895	3895	3895	3895	3895	3895	3895	3895	3895	3895	3895	3895	3895	3895
20.0	3770	3770	3770	3770	3770	3770	3770	3770	3770	3770	3770	3770	3770	3770
20.5	3607	3606	3606	3606	3606	3606	3606	3606	3607	3607	3607	3607	3607	3607
21.0	3408	3407	3407	3408	3408	3408	3408	3408	3408	3408	3408	3408	3408	3408
21.5	3180	3179	3179	3179	3179	3180	3180	3180	3180	3180	3180	3180	3180	3180
22.0	2917	2917	2917	2917	2917	2917	2917	2917	2917	2917	2917	2917	2917	2917
22.5	2645	2645	2645	2645	2645	2645	2645	2645	2645	2645	2645	2645	2645	2645
23.0	2363	2363	2363	2363	2363	2363	2363	2363	2363	2363	2363	2363	2363	2363
23.5	2070	2070	2070	2070	2070	2070	2070	2070	2070	2070	2070	2070	2070	2070
24.0	1782	1782	1782	1782	1782	1782	1782	1782	1782	1782	1782	1782	1782	1782
24.5	1561	1561	1561	1561	1561	1561	1561	1561	1561	1561	1561	1561	1561	1561

Table 5-1: Power curve, Power Optimized Mode PO2/PO2-0S

5.2 Ct Values, Power Optimized Mode PO2/PO2-0S

Wind	Air density kg/m ³													
	1.225	0.950	0.975	1.000	1.025	1.050	1.075	1.100	1.125	1.150	1.175	1.200	1.250	1.275
3.0	0.886	0.89	0.89	0.89	0.889	0.889	0.888	0.888	0.888	0.887	0.887	0.886	0.886	0.885
3.5	0.845	0.851	0.85	0.85	0.849	0.848	0.848	0.847	0.847	0.846	0.846	0.846	0.845	0.845
4.0	0.828	0.827	0.827	0.827	0.828	0.828	0.828	0.828	0.828	0.828	0.828	0.828	0.828	0.828
4.5	0.825	0.826	0.826	0.826	0.826	0.826	0.826	0.826	0.825	0.825	0.825	0.825	0.824	0.824
5.0	0.82	0.824	0.823	0.823	0.823	0.822	0.822	0.822	0.821	0.821	0.821	0.82	0.82	0.819
5.5	0.817	0.821	0.82	0.82	0.82	0.819	0.819	0.819	0.818	0.818	0.818	0.817	0.817	0.816
6.0	0.813	0.818	0.817	0.817	0.816	0.816	0.816	0.815	0.815	0.814	0.814	0.813	0.812	0.812
6.5	0.81	0.815	0.815	0.814	0.814	0.813	0.813	0.812	0.812	0.811	0.811	0.81	0.809	0.808
7.0	0.806	0.814	0.813	0.813	0.812	0.811	0.811	0.81	0.809	0.809	0.808	0.807	0.806	0.805
7.5	0.801	0.811	0.81	0.809	0.808	0.808	0.807	0.806	0.805	0.804	0.803	0.802	0.8	0.799
8.0	0.794	0.806	0.805	0.804	0.803	0.802	0.801	0.8	0.799	0.797	0.796	0.795	0.793	0.792
8.5	0.787	0.805	0.804	0.803	0.801	0.8	0.798	0.797	0.795	0.793	0.791	0.789	0.784	0.782
9.0	0.749	0.797	0.793	0.79	0.786	0.783	0.778	0.774	0.769	0.765	0.759	0.754	0.742	0.736
9.5	0.666	0.755	0.749	0.742	0.736	0.729	0.721	0.712	0.704	0.696	0.686	0.676	0.656	0.645
10.0	0.571	0.695	0.685	0.675	0.665	0.655	0.643	0.631	0.62	0.608	0.595	0.583	0.558	0.546
10.5	0.479	0.622	0.609	0.595	0.582	0.569	0.556	0.542	0.529	0.515	0.503	0.491	0.468	0.457
11.0	0.405	0.544	0.53	0.515	0.5	0.486	0.473	0.461	0.448	0.436	0.425	0.415	0.395	0.386
11.5	0.346	0.465	0.452	0.439	0.426	0.413	0.402	0.392	0.382	0.371	0.363	0.354	0.338	0.331
12.0	0.299	0.398	0.387	0.376	0.366	0.355	0.346	0.337	0.329	0.32	0.313	0.306	0.293	0.286
12.5	0.261	0.345	0.336	0.326	0.317	0.308	0.301	0.294	0.286	0.279	0.273	0.267	0.256	0.251
13.0	0.231	0.301	0.294	0.286	0.278	0.27	0.264	0.258	0.252	0.246	0.241	0.236	0.226	0.221
13.5	0.205	0.266	0.259	0.253	0.246	0.239	0.234	0.229	0.224	0.218	0.214	0.209	0.201	0.197
14.0	0.183	0.237	0.231	0.225	0.219	0.213	0.209	0.204	0.199	0.195	0.191	0.187	0.18	0.176
14.5	0.165	0.212	0.207	0.201	0.196	0.191	0.187	0.183	0.179	0.175	0.171	0.168	0.162	0.158
15.0	0.149	0.19	0.186	0.181	0.177	0.172	0.169	0.165	0.161	0.158	0.155	0.152	0.146	0.143
15.5	0.135	0.172	0.168	0.164	0.16	0.156	0.153	0.149	0.146	0.143	0.14	0.138	0.132	0.13
16.0	0.123	0.156	0.152	0.149	0.145	0.142	0.139	0.136	0.133	0.13	0.128	0.125	0.121	0.118
16.5	0.112	0.142	0.139	0.136	0.133	0.129	0.127	0.124	0.121	0.119	0.117	0.114	0.11	0.108
17.0	0.103	0.13	0.127	0.124	0.121	0.118	0.116	0.114	0.111	0.109	0.107	0.105	0.101	0.099
17.5	0.095	0.12	0.117	0.115	0.112	0.109	0.107	0.105	0.103	0.101	0.099	0.097	0.094	0.092
18.0	0.087	0.109	0.107	0.104	0.102	0.099	0.097	0.095	0.094	0.092	0.09	0.088	0.085	0.084
18.5	0.078	0.098	0.096	0.093	0.091	0.089	0.088	0.086	0.084	0.082	0.081	0.08	0.077	0.076
19.0	0.07	0.088	0.086	0.084	0.082	0.08	0.079	0.077	0.076	0.074	0.073	0.072	0.069	0.068
19.5	0.064	0.08	0.078	0.076	0.074	0.073	0.071	0.07	0.069	0.067	0.066	0.065	0.063	0.062
20.0	0.058	0.072	0.07	0.069	0.067	0.066	0.065	0.063	0.062	0.061	0.06	0.059	0.057	0.056
20.5	0.052	0.064	0.063	0.062	0.06	0.059	0.058	0.057	0.056	0.055	0.054	0.053	0.051	0.05
21.0	0.046	0.057	0.056	0.055	0.053	0.052	0.051	0.05	0.049	0.049	0.048	0.047	0.045	0.045
21.5	0.041	0.05	0.049	0.048	0.047	0.046	0.045	0.044	0.044	0.043	0.042	0.041	0.04	0.039
22.0	0.036	0.044	0.043	0.042	0.041	0.04	0.039	0.039	0.038	0.037	0.037	0.036	0.035	0.034
22.5	0.031	0.038	0.037	0.036	0.035	0.035	0.034	0.033	0.033	0.032	0.032	0.031	0.03	0.03
23.0	0.027	0.032	0.032	0.031	0.03	0.03	0.029	0.029	0.028	0.028	0.027	0.027	0.026	0.026
23.5	0.023	0.027	0.027	0.026	0.026	0.025	0.025	0.024	0.024	0.024	0.023	0.023	0.022	0.022
24.0	0.019	0.023	0.022	0.022	0.022	0.021	0.021	0.021	0.02	0.02	0.02	0.019	0.019	0.019
24.5	0.017	0.02	0.019	0.019	0.019	0.018	0.018	0.018	0.018	0.017	0.017	0.017	0.016	0.016

Table 5-2: C_t values, Power Optimized Mode PO2/PO2-0S

5.3 Sound Curves, Power Optimized Mode PO2/PO2-0S

Sound Power Level at Hub Height		
Conditions for Sound Power Level:	Measurement standard IEC 61400-11 ed. 3 Maximum turbulence at hub height: 30% Inflow angle (vertical): $0 \pm 2^\circ$ Air density: 1.225 kg/m^3	
Wind speed at hub height [m/s]	Sound Power Level at Hub Height [dBA] Power Optimized Mode P02 (Blades with serrated trailing edge)	Sound Power Level at Hub Height [dBA] Power Optimized Mode P02-0S (Blades without serrated trailing edge)
3	91.1	93.0
4	91.3	93.6
5	93.2	96.7
6	96.4	100.1
7	100.0	103.4
8	103.3	106.3
9	105.0	107.8
10	105.0	107.8
11	105.0	107.8
12	105.0	107.8
13	105.0	107.8
14	105.0	107.8
15	105.0	107.8
16	105.0	107.8
17	105.0	107.8
18	105.0	107.8
19	105.0	107.8
20	105.0	107.8

Table 5-3: Sound curves, Power Optimized Mode PO2/PO2-0S